

Project Proposal: Lake Geneva Water Temperature Forecast

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Repository (public): <https://github.com/mariamicioni/lake-geneva-temp>

1 Problem statement

Predict tomorrow's mean water temperature of Lake Geneva at Geneva (BAFU station 2606), one day ahead, targeting at least 20% lower MAE than persistence (persistence MAE 0.24 °C in October–January, so a target of at most 0.19 °C).

I want to predict the daily mean water temperature of Lake Geneva for the next day, measured where the lake flows out into the Rhône in Geneva (BAFU station 2606, Genève Halle de l'Île), about 1 km from the Bains des Pâquis. The forecast runs every evening at 18:00 and predicts the mean of the following calendar day, so the horizon is one day (6 to 30 hours ahead), in time to plan the next day. The scope is one station with daily values, evaluated from October 2026 to January 2027.

The forecast is meant for people who swim or paddle in the lake, which many do all year round. The official BAFU measurements only show the current temperature, not tomorrow's.

Success criterion: model MAE at least 20% lower than persistence on the out-of-time test set. On ten years of BAFU daily means (2016–2026), persistence has an MAE of 0.52 °C over all seasons but only 0.24 °C in October–January, when the lake cools slowly. For my evaluation period the target is therefore an MAE of at most 0.19 °C.

2 Originality & motivation

I am attached to the Lake Geneva region for family reasons, and there people swim and do sports like stand-up paddling on the lake all year round. The water temperature is one of the first things you check before going in, so knowing it for the next day is useful. The problem is also interesting for ML: on calm days the temperature hardly changes and persistence is hard to beat, but after strong wind or a change in the weather the surface water gets mixed, and the temperature at the outflow can change by more than 1 °C in a day.

I went through the 15 FS26 projects on mlops-lab.ch [7] and the KTH ID2223 project lists of 2024 (31 projects) and 2025 (35 projects) [8]. None of them predicts water temperature. The closest one is "Predict UK river flooding events" (ID2223, 2024), which classifies floods in the UK. Eawag's Alplakes platform [6] already publishes a physics-based simulation of Lake Geneva. My project is different because it is a data-driven model for one measured point that is evaluated against a baseline and retrained automatically; a comparison with Alplakes is an optional extra.

3 Data source & features

The water data comes from the Federal Office for the Environment (FOEN/BAFU) [1]. Station 2606 measures water temperature and discharge every 10 minutes. For live ingestion I use the REST API of existenz.ch [2], with the official LINDAS feed as fallback [3]. For historical water backfill I use the official BAFU Data Platform GraphQL API [1]. The weather comes from the Open-Meteo Forecast API [4] for Geneva (46.20 N / 6.14 E). For backfill and backtesting I use Open-Meteo's Single Runs API [5], selecting a run that was actually available by 18:00, accounting for model publication delay.

The existenz.ch REST history covers only about 32 days, but the BAFU Data Platform provides station 2606 history back to 2016 (daily, hourly and 10-minute values), so training is not limited to the poller start. Single Runs exist from mid-April 2026, giving about 170 training days today plus one per day; the Previous Runs API (two-day-old forecasts, back to 2022) allows a longer backfill. A daily GitHub Actions job saves new raw CSVs in a GitHub Release (outside Git) until the GCS bucket exists; from MS2 raw and feature data are stored in GCS and tracked with DVC. I start with a regularised linear model (Ridge).

The label is the mean of the measured temperatures at station 2606 on the next day. It is only used if at least 120 of the 144 readings exist (83.3% completeness), which tolerates short gaps while rejecting substantially incomplete days; I will check this threshold during MS2. It is a measurement from the future, so it cannot be calculated from the features. There is no rare class because this is a regression, but I will report the error separately for days where the temperature changes by more than 0.8 °C, because those are the days where persistence fails.

Features per day:

- water_mean_d0: mean water temperature today (until 18:00)
- water_last: last reading before 18:00
- water_mean_3d, water_trend_3d: 3-day statistics using only information available by 18:00; mean over the current partial day plus the previous two days, and change versus three days earlier
- flow_mean_d0: mean discharge today
- air_temp_fc_d1, sunshine_fc_d1, wind_max_fc_d1: Open-Meteo forecast for tomorrow (mean air temperature, sunshine duration, maximum wind speed)

Leakage: I split by time and never shuffle randomly. During development I use expanding-window validation; the final evaluation uses the latest period as an out-of-time holdout. Only readings and weather forecasts actually available before 18:00 are used. Backfill respects forecast publication delay. All daily values use Europe/Zurich time, and training and inference use the same weather model and feature code.

4 System design

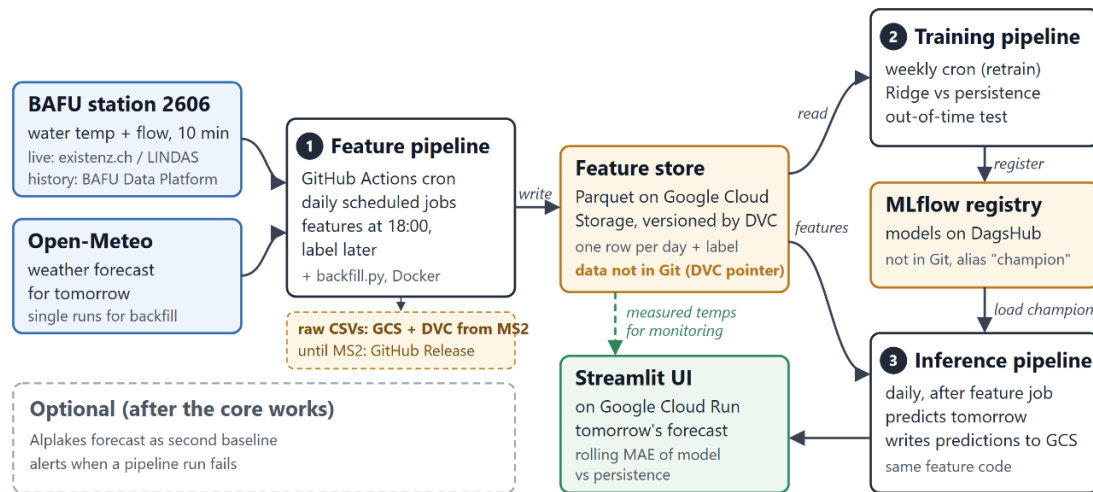


Figure 1: FTI architecture; numbered boxes are the three pipelines, the dashed box is optional.

Component	Tool	Why
Feature store	Parquet files on Google Cloud Storage, versioned with DVC [9]	One row per day is very little data, so no separate feature store service is needed. Parquet snapshots live in GCS and are tracked with DVC; the DVC pointer files are committed to Git. Paid with the GCP student credits.
Orchestration	GitHub Actions (cron)	Free and part of the repo. Runs the daily feature and inference jobs and the weekly retraining.
Experiment tracking and registry	MLflow, hosted on DagsHub [10]	Free hosted MLflow for public repos, so I don't run my own server. Logs parameters, metrics and the model; the registry alias champion marks the model that is served.
Model	scikit-learn (Ridge)	Small tabular data set; a simple model is fast to train and easy to explain.
Serving and UI	Streamlit on Google Cloud Run	Same cloud as the feature store, paid with the GCP student credits, scales to zero when unused. Shows tomorrow's forecast and model performance versus persistence over time.
Packaging	Docker (multi-stage, non-root user), uv (uv.lock)	The same environment locally, in GitHub Actions and on Cloud Run, with pinned dependencies. Secrets only as GitHub Secrets, never in the image.

Reproducibility: data and models are never stored directly in Git [11]; Git holds code, configuration and small DVC pointer files. Every training run logs its Git commit, DVC data version, parameters, metrics and model, so each result can be traced back to its code and data.

References and AI use

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- [11] Bär, Florian (2026). Versioning and Containerization, MLOps HS26 Block 1, lecture slides. Lucerne University of Applied Sciences and Arts.

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